Medical Appointment No-show Analysis Report

Project Overview

This project analyzes the Medical Appointment No Shows dataset from Kaggle to identify factors associated with missed medical appointments. The goal is to explore attendance patterns, identify possible reasons for patient no-shows, and provide practical recommendations based on the analysis.

Dataset Information

- **Source:** Kaggle – Medical Appointment No Shows Dataset
- **Number of Records:** 110,527
- **Number of Features:** 14

The dataset contains demographic information, appointment details, health conditions, and attendance status for each patient.

Data Cleaning

The following preprocessing steps were completed before starting the analysis:

- Converted appointment dates into datetime format.
- Removed one invalid age value (-1).
- Verified that there were no missing values.
- Verified that there were no duplicate records.
- Created a new feature called **WaitingDays**.
- Grouped waiting days into meaningful categories using **WaitingGroup**.

Exploratory Data Analysis

1. Overall No-show Rate

	proportion
No-show	
No	79.81
Yes	20.19

Observation

Approximately 20% of all appointments resulted in patient no-shows.

Insight

Most patients attended their appointments, but the no-show rate remains high enough to affect healthcare efficiency.

Recommendation

Continue monitoring attendance rates and identify patients with higher no-show risk.

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2. Gender Analysis

Observation

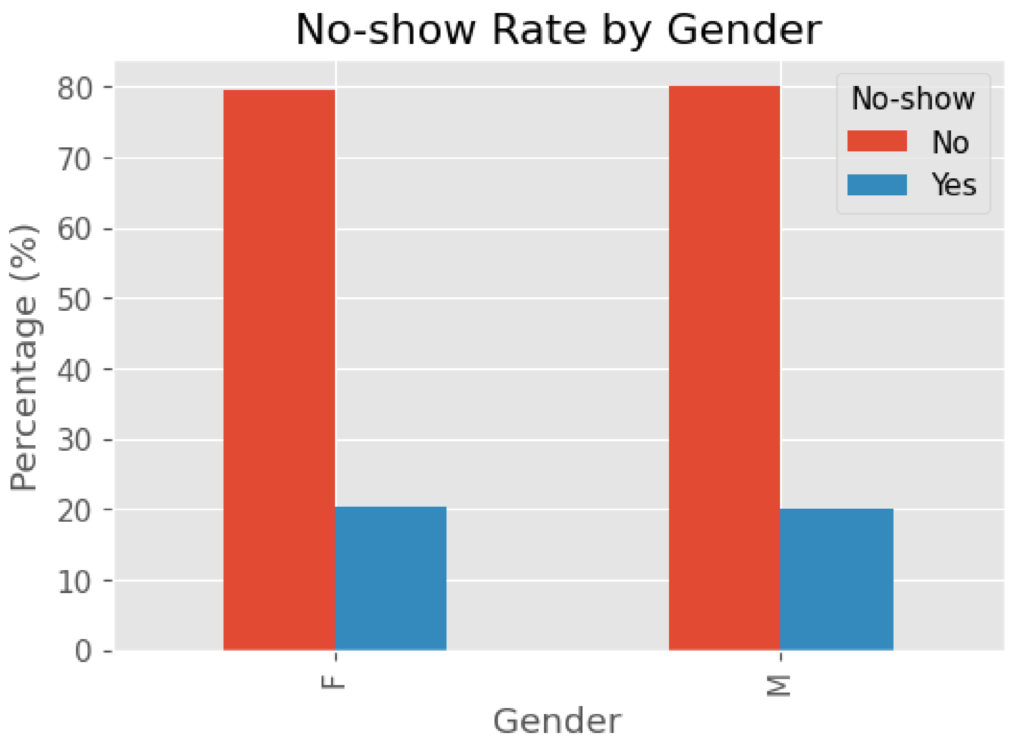
Attendance rates were very similar between males and females.

Insight

Gender showed little association with appointment attendance.

Recommendation

Gender should not be considered a primary factor when predicting no-shows.



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3. SMS Reminder Analysis

Observation

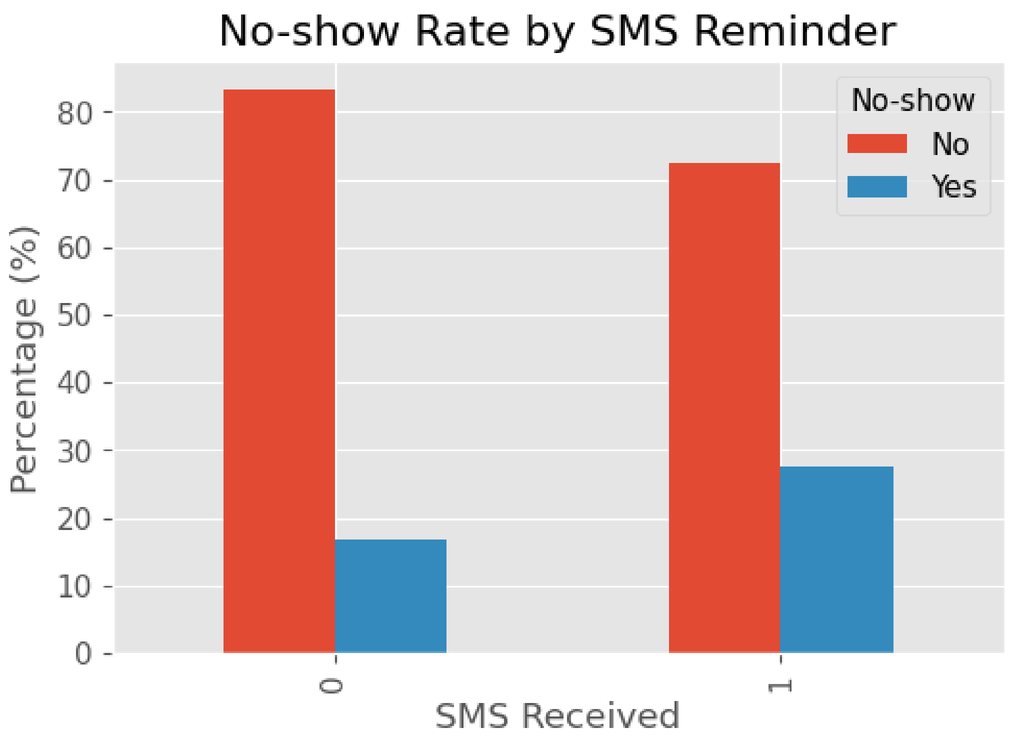
Patients who received SMS reminders had a slightly higher no-show rate than those who did not.

Insight

Receiving an SMS reminder alone does not appear to improve attendance in this dataset.

Recommendation

Review reminder timing and communication strategies instead of relying only on SMS notifications.



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4. Waiting Time Analysis

Observation

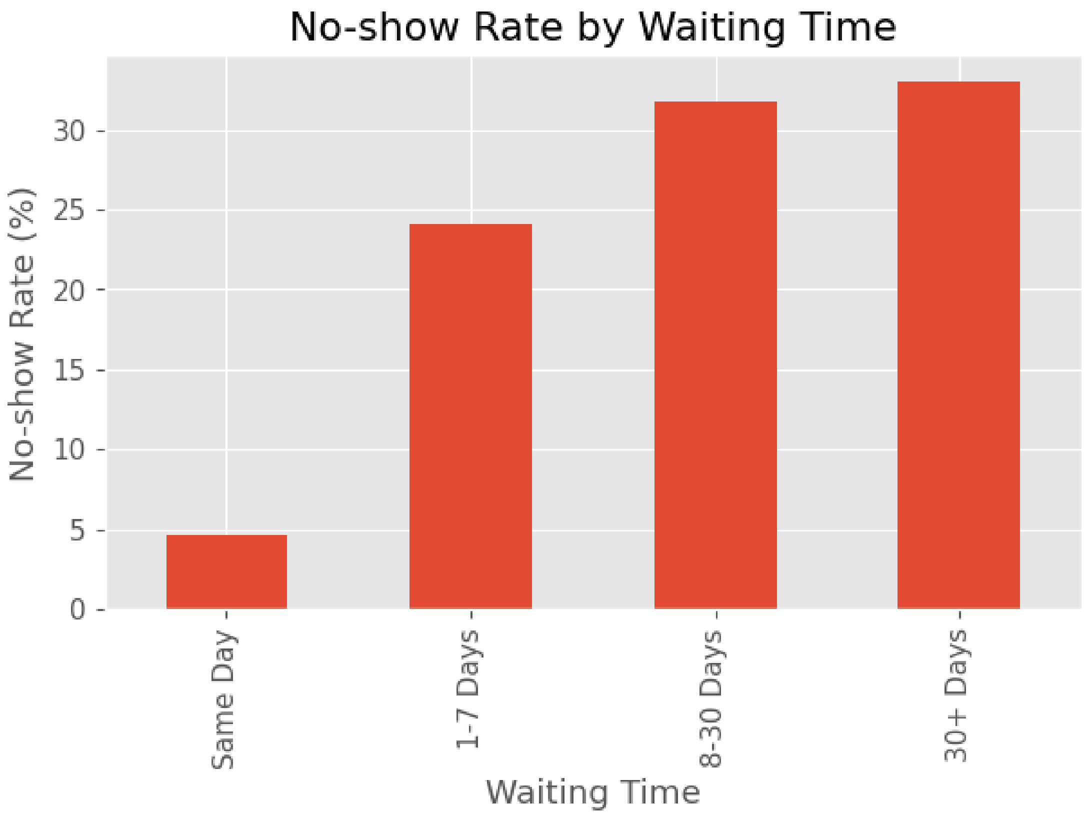
Patients with longer waiting times were more likely to miss their appointments.

Insight

Waiting time was the strongest factor associated with appointment attendance.

Recommendation

Reducing waiting times may significantly improve attendance rates.



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5. Age Analysis

Observation

Attendance patterns were generally similar across age groups, with only small differences.

Insight

Age alone was not a strong predictor of appointment attendance.

Recommendation

Age should be combined with other variables when analyzing patient behavior.

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6. Diabetes Analysis

Observation

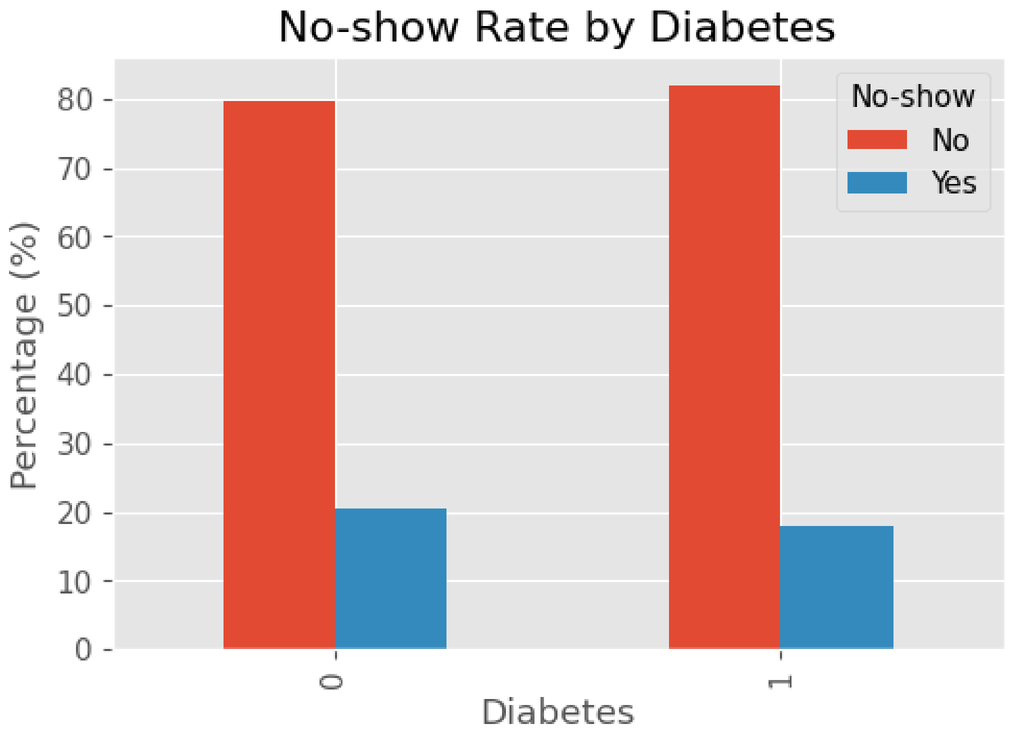
Patients with and without diabetes showed very similar attendance rates.

Insight

Diabetes status had only a limited association with appointment attendance.

Recommendation

Additional patient characteristics should be considered before making decisions.



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7. Scholarship Analysis

Observation

Patients receiving scholarships had a slightly higher no-show rate than those without scholarships.

Insight

The difference was relatively small and may be influenced by other socioeconomic factors.

Recommendation

Further analysis is recommended before drawing conclusions.

Key Findings

- Around one in five appointments resulted in a no-show.
- Waiting time showed the strongest association with missed appointments.
- Same-day appointments achieved the highest attendance rates.
- Gender, age, diabetes, and scholarship status showed only minor differences.
- SMS reminders alone were not associated with improved attendance.

Recommendations

- Reduce appointment waiting times whenever possible.
- Monitor patients with long waiting periods.
- Improve reminder strategies beyond SMS notifications.
- Continue using data analysis to support scheduling decisions.

Conclusion

This project demonstrates how healthcare data can be transformed into actionable insights through data cleaning, exploratory analysis, and visualization. The findings highlight the importance of waiting time as a potential factor influencing appointment attendance and illustrate how data analysis can support better operational decision-making in healthcare.